

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CASE STUDY

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	20% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	S.Mahammed Nayeem	Reg. No.:	192425294
Section / Batch:	AI &ML/2024	Date:	06-08-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues, provide an in-depth analysis, and propose innovative, feasible solutions with strong justification.

Question Type	Focus Area	Questions
Case Study	Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication	Q1

SECTION A – CASE STUDY

Code of Conduct

I S.Mahammed Nayeem / 192425294(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S.Mahammed Nayeem

PYTHON CODE :

```
def binary_search(words, meanings, key):
    left = 0
    right = len(words) - 1
    key = key.lower()

    while left <= right:
        mid = (left + right) // 2

        if words[mid].lower() == key:
            return meanings[mid]
        elif words[mid].lower() < key:
            left = mid + 1
        else:
            right = mid - 1

    return "Word not found"

words = ["apple", "cat", "dog", "elephant", "fish", "goat"]
meanings = [
    "A fruit",
    "A small domestic animal",
    "A loyal domestic animal",
    "A large animal",
    "An aquatic animal",
    "A farm animal"
]

key = input().strip()

result = binary_search(words, meanings, key)
print("Meaning:", result)
```

OUTPUT :

QUESTIONS

Q1

Digital Dictionary Lookup
System (Binary Search)
100 marks

1/1 answered

Q1 Digital Dictionary Lookup System (Binary Search)

100 marks

A digital dictionary stores words in sorted order and must retrieve meanings quickly. Analyze how Binary Search improves lookup efficiency. Identify requirements such as sorted structure and quick access. Apply divide and conquer principles to explain searching. Analyze time complexity and performance. Design a solution and justify why Binary Search is optimal.

Your Code

```
1 def binary_search(words, meanings, key):
2     """
3     words: list of sorted words
4     meanings: list of meanings corresponding to words (same length)
5     key: word to search
6     returns: meaning if found, else None
7     """
8     left, right = 0, len(words) - 1
9     key = key.strip().lower()
10
11     while left <= right:
12         mid = (left + right) // 2
13         mid_word = words[mid].strip().lower()
14
15         if mid_word == key:
16             return meanings[mid]
17         elif mid_word < key:
18             left = mid + 1
19         else:
20             right = mid - 1
21
22     return None
23
24
25 def main():
26     words = ["apple", "cat", "dog", "elephant", "fish", "goat"]
27     meanings = [
```

Input

stdin input

Output

Meaning: a large animal

RUBRICS FOR CALCULATION

Case Study					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25%	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20%	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20%	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10%	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%				
Application of Concepts	25%				
Analysis	20%				

Solution Design	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____